
Predict, Optimize, Respond: An Integrated Framework for Sudden River Pollution

Summary

To address the emergency response needs for sudden water pollution incidents in the Chop-tank River Basin, this paper constructs an **integrated emergency management framework** that combines hydrological analysis, transport simulation, decision optimization, and monitoring design. By integrating physical mechanisms with engineering practice, this framework establishes a complete technical chain from pollution prediction to scientific decision-making.

For Problem 1, based on hourly discharge data for June 2025 and historical statistical records, we performed a flow characteristic analysis using the **percentile classification method**. This accurately identified the study period as a **typical flow** scenario, providing a reliable foundation for subsequent modeling.

For Problem 2, we developed a **one-dimensional advection-dispersion-reaction equation model** for the 30-km river reach, employing the **differential evolution algorithm** for numerical solution. The simulation predicted that the pollutant would reach a **peak concentration of 0.47 mg/L at 117.3 hours**, and quantified the duration of exceedance above regulatory thresholds. Systematic parameter sensitivity analysis verified the robustness of the model under varying dispersion conditions.

For Problem 3, we constructed a **comprehensive objective function** incorporating shutdown costs, health risks, and water shortage penalties. Adopting a simulation-based optimization approach, we transformed the complex control problem into a **dual-threshold hysteresis controller**. Numerical optimization yielded **optimal shutdown and restart thresholds of 0.1584 mg/L and 0.1384 mg/L**, respectively. Subsequent scenario analysis demonstrated that this strategy effectively balances safety and economic considerations across different pollution scenarios.

For Problem 4, we developed a **sensor placement optimization model based on a five-dimensional scoring function**, using a **two-layer algorithm combining greedy selection and Monte Carlo validation** to determine the optimal sensor layout. The final three-tier deployment at **3 km, 8 km, and 18 km** upstream reduced the time to first detection from **77.18 hours to 2.37 hours**, providing nearly three days of advance warning and response window.

For Problem 5, we designed a **technology transfer plan for water utility management**, translating mathematical models into operational rules through plain-language explanations. Comparative analysis highlighted the advantages of the dual-threshold strategy over conventional methods, along with specific implementation recommendations.

The integrated emergency response framework established in this study achieves **closed-loop management from scientific prediction to engineering decision-making**, providing a systematic and operable solution for river pollution emergency management.

Keywords: one-dimensional advection-dispersion model, dual-threshold hysteresis control, Monte Carlo validation, greedy selection strategy

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1 Introduction

1.1 Background

Rivers are vital arteries for water supply, agriculture, and ecology. In the United States, thousands of water systems depend on river intakes to provide raw water for municipal use. However, these intakes face significant risks from upstream pollution events, such as chemical spills during or after storm events. A timely and scientifically informed operational response is essential to protect public health while minimizing economic disruption.

This study focuses on a hypothetical spill scenario along the Choptank River, a perennial stream in the Atlantic Coastal Plain of Maryland. The Choptank River near Greensboro, MD (USGS gauge 01491000) is characterized by a groundwater-dominated baseflow regime, supplemented by seasonal rainfall. Its hydrological behavior exhibits distinct wet and dry periods, influencing contaminant transport dynamics and operational decisions for downstream water intakes.

1.2 Literature Review

A robust decision framework for intake management must be grounded in river hydrology and contaminant transport. Prior studies in the Choptank River Basin provide essential insights that inform our modeling approach.

- **Hydrological Basis.** Seasonal evapotranspiration governs the river's flow regime, with peak discharges typically occurring in spring and low baseflows in late summer [4]. This seasonal variability strongly influences advective transport rates and therefore informs the flow-dependent structure of our contaminant transport model.
- **Storm-driven Contaminant Dynamics.** During rainfall events, particulate phosphorus and suspended solids exhibit sharp concentration increases on the rising limb of the hydrograph, while nitrate concentrations tend to decrease due to dilution effects [5]. These contrasting behaviors motivate the inclusion of a flow-dependent attenuation term in the transport model and highlight the critical importance of response timing for intake shutdown decisions.
- **Export Patterns.** More than 70% of annual phosphorus export and nearly all sediment transport occur during stormflow, whereas nitrogen export is distributed more evenly between baseflow and storm events [5]. This finding suggests that effective monitoring strategies must prioritize early detection of rapidly developing storm-driven contaminant plumes.
- **Groundwater Nitrate Inputs.** In addition to event-driven inputs, direct groundwater discharge can supply elevated nitrate concentrations, reaching up to 19.6 mg N L^{-1} , and represents a persistent background source to the river system [6]. Accordingly, our modeling framework accounts for both continuous background loading and episodic pulse inputs.

In summary, contaminant transport in the Choptank River is strongly flow-dependent and pollutant-specific. These insights guide the development of a dynamic advection–dispersion framework, a risk-informed intake shutdown strategy, and a sensor network design optimized for early detection of storm-mobilized contaminants.

1.3 Restatement of the Problem

This study addresses four tasks following a river contamination event, aiming to support reliable and economically sound intake operation under uncertainty.

1. Hydrograph Characterization & Contaminant Transport Modeling Using the provided hourly discharge data, we must first characterize the June 2025 flow regime as low, typical, or high relative to historical records. This characterization establishes the hydrological basis for a one-dimensional advection-dispersion model that must simulate contaminant transport over the 30 km reach. The model is required to predict the **arrival time, peak concentration, and exceedance duration** at the intake, with sensitivity to key transport parameters evaluated within physically realistic ranges.

2. Shutdown Decision Strategy An actionable, real-time operational rule must be designed to determine **when to shut down, for how long, and when to reopen**. This rule should be derived from an explicit objective function that balances **shutdown costs, water-supply reliability, and water-quality risk**. The strategy must remain robust to uncertainties in both hydrological conditions and model parameters.

3. Optimal Early-Warning Sensor Network Design With a budget allowing at most three online concentration sensors, optimal locations must be selected from the provided candidate set, accounting for measurement noise and a 5-minute reporting delay. The placement should be justified by quantifying the network's improvement in **warning time** and reduction in **expected operational loss** compared to a no-sensor baseline.

4. Management Communication A one-page, non-technical memo must be prepared for utility management, clearly explaining the recommended intake operation strategy and sensor placement, and justifying their advantages over simpler alternatives such as fixed-threshold shutdowns or clustering all sensors at the intake.

1.4 Our Work

1. Data Preparation and Problem Setup

We first established key modeling assumptions to translate the real-world problem into a computable framework. We then processed and integrated multiple datasets, including historical hydrological records and river geometry data, to create a reliable foundation for all subsequent simulations.

2. Hydrological Analysis and Contaminant Transport Simulation

- (a) Applied a percentile-based statistical method to classify the baseline flow regime.
- (b) Developed a deterministic one-dimensional advection-dispersion-reaction model and employed a differential evolution algorithm for its numerical solution to simulate the downstream transport of the pollutant.

3. Shutdown Decision Strategy Design

- (a) Formulated the intake operation as a stochastic optimization problem, constructing a comprehensive objective function to quantify the trade-off between economic costs and public health risks.

- (b) Designed a practical dual-threshold hysteresis controller and determined its optimal parameters through simulation-based optimization, thereby translating the theoretical framework into an executable operating rule.

4. Early-Warning Sensor Network Optimization

- (a) Built a multi-criteria scoring function to evaluate potential sensor locations, incorporating factors such as warning-time gain and reliability.
- (b) Developed a two-layer robust optimization algorithm that combines a greedy selection strategy with Monte Carlo validation to solve the sensor placement problem under uncertainty.

5. Solution Translation and Management Communication

We synthesized the outputs of all mathematical models to form a clear, actionable emergency response plan. We also prepared a professional recommendation memorandum for water utility management, effectively bridging the gap between scientific modeling and operational decision-making.

Figure 1 visualizes the integrated workflow of our modeling approach.

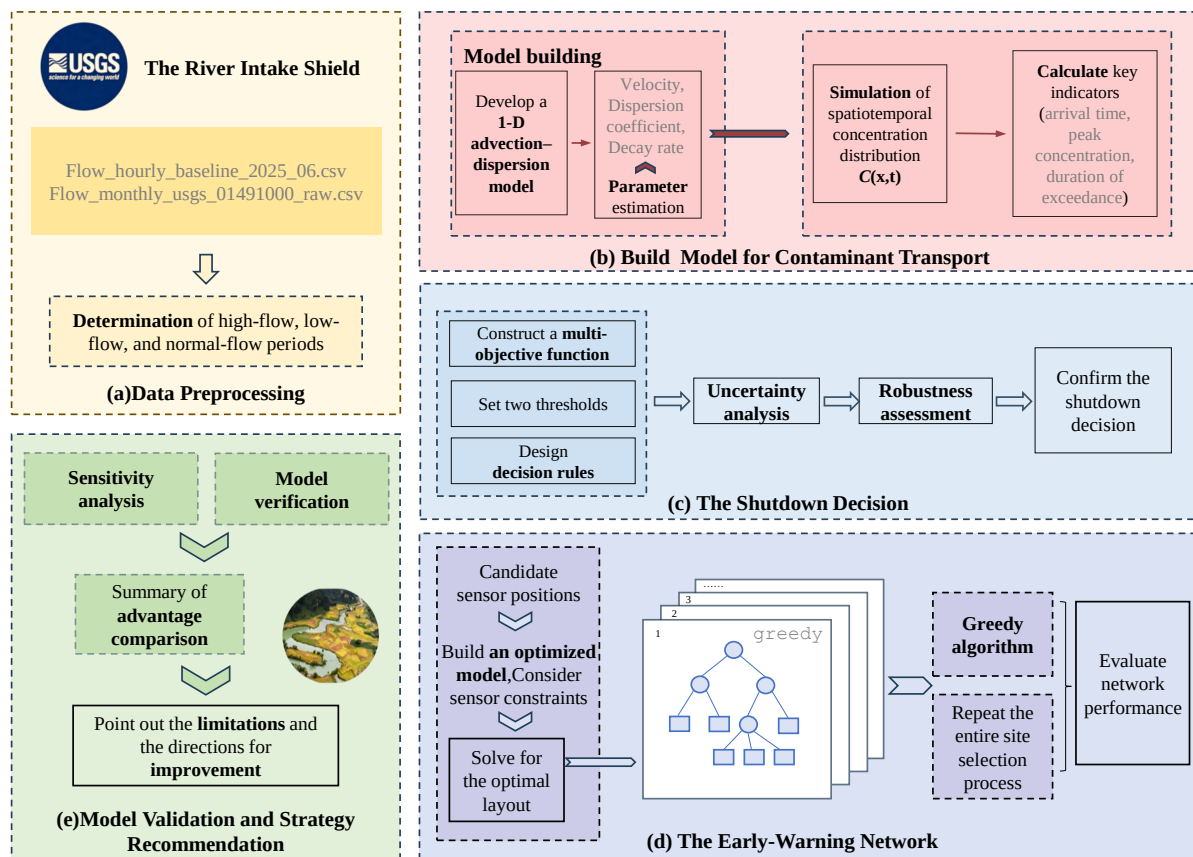


Figure 1: Model Framework

2 Assumptions and Justification

To simplify the problem and make it convenient for us to simulate real-life conditions, we make the following basic assumptions, each of which is properly justified.

Assumption 1 Pollutant transport follows one-dimensional advection-dispersion with complete cross-sectional mixing. Sedimentation, resuspension, and adsorption are neglected.

Justification: For dissolved contaminants over the simulation period, advection-dispersion dominates transport. The 30-km reach length justifies the 1D approximation, focusing on longitudinal transport to predict intake arrival times while neglecting cross-sectional gradients.

Assumption 2 Sensor noise follows a zero-mean normal distribution and is mutually independent.

Justification: This reflects typical sensor characteristics: zero mean eliminates systematic bias, normality handles random errors, and independence simplifies multi-sensor reliability analysis.

Assumption 3 Urban demand follows a 24-hour pattern and can be reduced by 60% during shutdowns.

Justification: This allows shortage cost quantification and aligns with practical emergency measures like water-conservation notices.

3 Notations

Table 1: Model Notation Summary

Symbol	Description	Unit	Symbol	Description	Unit
<i>Contaminant Transport (Model I)</i>					
$C(x, t)$	Conc. at x, t	mg/L	$Q(t)$	River discharge	m ³ /s
$C(t)$	Conc. at intake	mg/L	$v(t)$	Flow velocity	m/s
$A(x)$	Cross-sectional area	m ²	D	Dispersion coefficient	m ² /s
k	Decay rate	day ⁻¹	$S(x, t)$	Spill source term	mg/(L·s)
<i>Intake Operation (Model II)</i>					
$u(t)$	Intake status	—	c_{off}	Shutdown threshold	mg/L
c_{on}	Restart threshold	mg/L	C_{std}	Safety standard	mg/L
C_{mand}	Mandatory limit	mg/L	Q_{max}	Max intake capacity	m ³ /s
$Q_{\text{eff}}(u)$	Effective supply capacity	m ³ /s	$D(t)$	Urban demand	m ³ /s
$C_m(t)$	Measured conc. at intake	mg/L	λ_S	Shutdown cost coef.	CNY/h
λ_R	Risk cost coef.	CNY/(mg·L ⁻¹ ·h)	λ_D	Shortage penalty coef.	CNY/m ³
J	Total expected cost	CNY			
<i>Sensor Network (Model III)</i>					
$\mathcal{L}$	Candidate locations set	km	S_{sen}	Selected network	—
ΔT	Warning time gain	h	SNR	Signal-to-noise ratio	—
W	Detection-window length	h	ρ	Spatial correlation	—
R	Detection reliability	—	σ	Noise std. dev.	mg/L
τ	Reporting delay	min			

4 Hydrological Data Characterization

4.1 Hydrologic Baseline for June 2025

The baseline hourly river discharge for June 2025 (`flow_hourly_baseline_2025_06.csv`) is used directly as the primary input $Q(t)$ for all contaminant transport simulations. This hourly hydrograph represents the flow conditions assumed during the spill event and is applied without temporal aggregation or reconstruction. Historical monthly discharge statistics from the Choptank River at Greensboro, MD (USGS 01491000) provide hydrological context and are used solely to classify the June 2025 condition as low-, typical-, or high-flow; they do not modify the hourly discharge series.

4.2 Statistical Characteristics of June Discharge

Analysis of historical June flows reveals substantial interannual variability, with generally low median flows punctuated by episodic high-discharge years associated with storm events, as illustrated in Figure 2. The empirical distribution is strictly positive and strongly right-skewed, violating the support and symmetry assumptions of the normal distribution. Gamma-type distributions, defined on the positive real line and capable of modeling skewness, therefore offer a more appropriate representation of observed flow behavior.

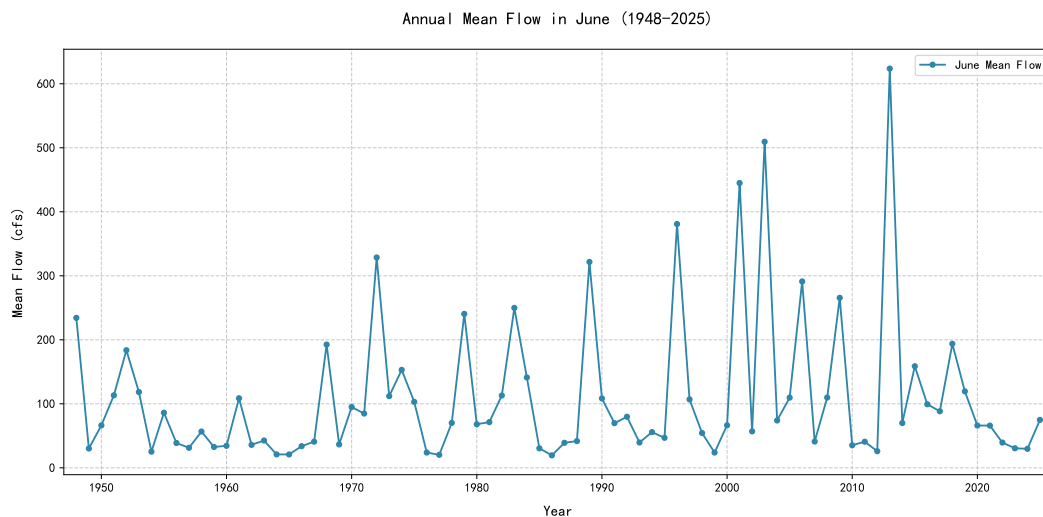


Figure 2: Annual Mean Flow in June (1948-2025)

4.3 Percentile-Based Flow Regime Classification

We analyzed the 78-year record of June average discharges (1948–2025) using two distinct but related methods to contextualize the June 2025 flow condition: ordinal ranking and percentile classification.

Firstly, we performed a simple ordinal ranking by arranging all 78 years of June average discharges in ascending order. In this ranked list, the June 2025 value of 114.1 cfs was found to occupy a position equivalent to approximately the 75th percentile rank.

Secondly, to align with standard hydrological classification, we applied a percentile-based regime analysis using the empirical cumulative distribution function (CDF). Based on the CDF, we categorized June flows into three regimes:

- **Low-flow:** below the 25th percentile,
- **Typical-flow:** 25th–75th percentile,
- **High-flow:** above the 75th percentile.

Figure 3 provides a detailed view of this distribution-based classification. The June 2025 average discharge of 114.1 cfs corresponds to the 62nd percentile (CDF ≈ 0.624), placing it definitively **within the Typical-flow regime** and confirming statistically typical conditions.

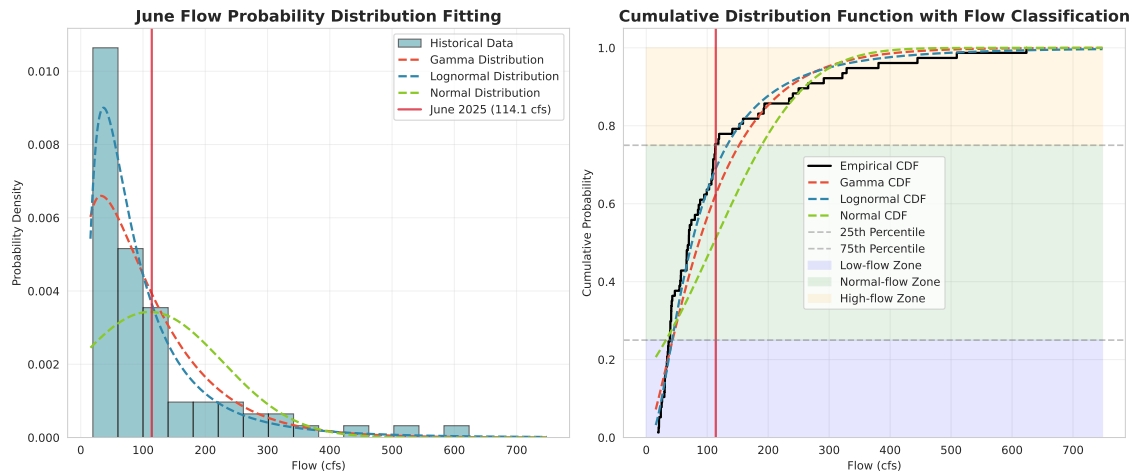


Figure 3: June Streamflow Distribution and Flow Regime Classification

The difference between the $\sim 75^{\text{th}}$ ordinal rank and the 62.4th percentile (CDF) is methodological. The percentile rank is a discrete count of positions, while the CDF percentile uses interpolation for a continuous distribution, providing a more precise statistical location. Crucially, both results lead to the same operational conclusion: June 2025 discharge is above the historical median but within the typical range.

These diagnostics justify the use of the provided June 2025 hourly hydrograph as a representative baseline for contaminant transport modeling, while acknowledging that even typical flows can be punctuated by episodic storms.

5 Model I : Contaminant Transport Model

5.1 Model Overview

We developed a one-dimensional advection–dispersion (ADR) model to simulate how the pollutant plume moves downstream over the 30 km river reach. This model accounts for the changing river flow and the natural spreading of the plume. Using baseline June 2025 flow data, the simulation successfully predicted the arrival time, peak concentration, and how long the pollutant would exceed safety thresholds at the intake. Through sensitivity analysis, we tested the dispersion coefficient across its realistic range and confirmed that it is the most critical parameter, ensuring our predictions remain robust under varying river conditions.

5.2 Model Establishment

To predict the downstream transport of contaminants following a spill event, a one-dimensional advection-dispersion-reaction (ADR) model is developed to describe the temporal and spatial evolution of pollutant concentration along the river. Given the relatively long study reach and sufficient lateral and vertical mixing at the time scale of interest, pollutant concentration is assumed to be uniform across each river cross-section, and transport is modeled only along the longitudinal direction of the river.

Let $C(x, t)$ denote the pollutant concentration at downstream distance x and time t . The transport process is governed by:

$$\frac{\partial C}{\partial t} + v(t) \frac{\partial C}{\partial x} = D \frac{\partial^2 C}{\partial x^2} - kC + S(x, t) \quad (1)$$

where the terms represent advection by river flow, longitudinal dispersion, first-order decay, and external contaminant input, respectively.

The river flow velocity is computed directly from the provided discharge data as:

$$v(t) = \frac{Q(t)}{A(x)} \quad (2)$$

where $Q(t)$ is the hourly discharge time series for June 2025, used as the baseline hydrologic input for all simulations. The cross-sectional area A is obtained from the river geometry data (`river_geometry_segments.csv`) and is treated as constant within each river segment, allowing the flow velocity to vary over time in response to changing discharge conditions.

It is calculated as the product of the provided width and depth data. This variation along the longitudinal distance x (in km) from the downstream reference point is described by the following piecewise function:

$$A(x) = \begin{cases} 39.6, & 0 \leq x < 10, \\ 50.0, & 10 \leq x < 20, \\ 61.6, & 20 \leq x \leq 30. \end{cases} \quad (3)$$

The longitudinal dispersion coefficient is set to $D = 50.0 \text{ m}^2/\text{s}$, consistent with the parameter range provided in the problem statement. To assess model robustness, sensitivity analyses are conducted over the range $10.0 \leq D \leq 200.0 \text{ m}^2/\text{s}$. The model adopts the provided first-order decay rate $k = 0.05 \text{ day}^{-1}$ to represent realistic attenuation processes during transport, establishing it as the baseline scenario for all primary analyses. To ensure a rigorous upper bound for risk assessment, a conservative scenario with zero decay ($k = 0$) is also defined and evaluated separately as a worst-case benchmark.

The contaminant spill is represented by a source term $S(x, t)$, which is assumed to be spatially localized at the spill location and temporally limited in duration. This formulation allows both instantaneous and short-term continuous release scenarios to be modeled. The resulting concentration time series at downstream locations provide the basis for subsequent analysis of pollutant exposure and system response.

5.3 Results

Based on the developed one-dimensional advection-dispersion-reaction (ADR) model, we simulated the downstream transport of the contaminant plume over the 30 km reach under baseline June 2025 flow conditions. The spatiotemporal evolution of concentration is shown in Figure 4.

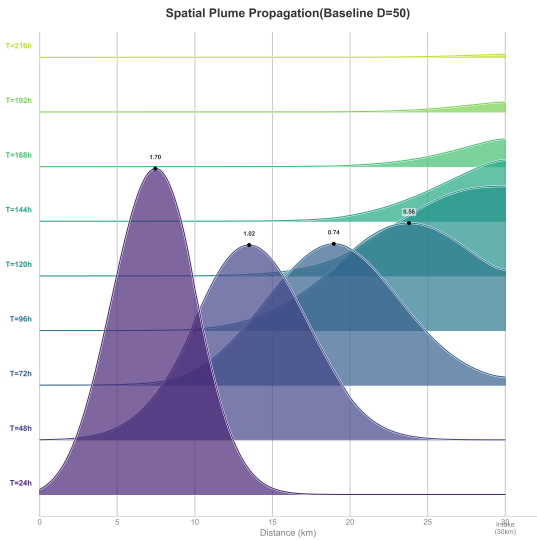


Figure 4: Contaminant Plume Evolution

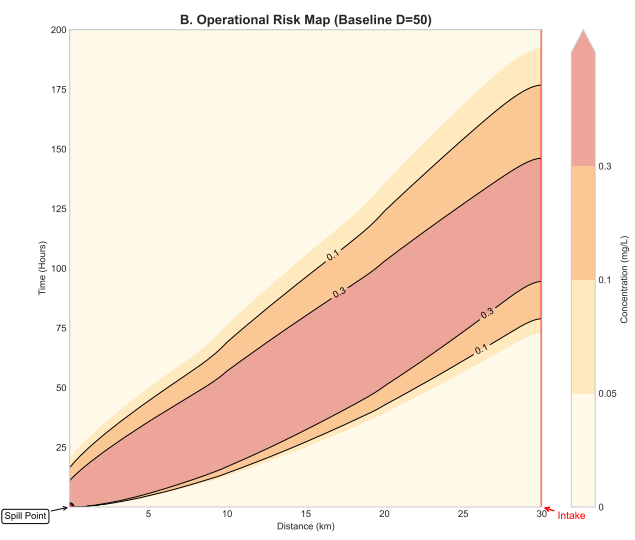


Figure 5: Operational Risk Assessment

The simulation indicates that the contaminant plume travels steadily downstream with the river flow, while longitudinal dispersion causes gradual spreading and attenuation of the concentration peak. The plume reaches the intake location with a clearly defined rising phase, peak, and recession, allowing key exposure metrics to be quantified.

Under the baseline dispersion coefficient $D = 50 \text{ m}^2/\text{s}$, key exposure metrics at the intake are summarized in Table 2:

Table 2: Critical Time Nodes for Pollutant Transport ($D = 50 \text{ m}^2/\text{s}$)			
Critical Node	Time (h)	Concentration (mg/L)	Duration (h)
Exceeding warning threshold	78.9–176.8	> 0.1	97.9
Exceeding shutdown threshold	94.5–146.1	> 0.3	51.6
Peak concentration	117.3	0.47	—

These results directly answer the problem requirements by quantifying the arrival time, peak concentration, and duration above regulatory thresholds at the intake under realistic flow conditions.

Figure 5 further visualizes the joint evolution of concentration and risk level over time, clearly indicating the high-risk period during which intake operation would pose unacceptable water-quality risks. This visualization bridges the physical transport results with subsequent operational decision-making.

6 Model II: Intake Shutdown Strategy

6.1 Model Overview

We developed a dynamic shutdown decision model to minimize the expected total cost comprising shutdown expenses, health risks, and water shortage penalties. Given the computational complexity of real-time optimization, we simplified this theoretical framework into a dual-threshold hysteresis controller, whose optimal parameters were determined through numerical optimization: a shutdown threshold of 0.1584 mg/L and a restart threshold of 0.1384 mg/L. Simulation under the baseline scenario confirms that this rule maintains pollutant concentrations below the safety limit throughout the event. Further robustness tests across multiple pollution scenarios ($D = 10, 50, 200$) demonstrate that the strategy adapts effectively to varying contamination patterns, exhibiting both stability and environmental adaptability while ensuring water safety.

6.2 Model Establishment

6.2.1 Model Formulation for the Shutdown Decision

When a pollutant spills upstream, operating the water intake becomes a balancing act. On one hand, we must keep the water flowing to avoid costly shutdowns and potential shortages. On the other hand, we must ensure the water is safe to drink. To handle this trade-off systematically, we frame the problem as an optimal control task. Our goal is to find an on-off schedule $u(t)$ for the intake that minimizes the expected total cost, where $u(t) = 1$ means the intake is open and $u(t) = 0$ means it is closed.

We break down the total cost into three parts, each capturing a different economic impact.

First, shutting down the intake costs money directly. We model this as a cost that adds up linearly with the total time the intake is off:

$$C_S(u(t)) = \lambda_S \int_0^T (1 - u(t)) dt \quad (4)$$

Here, $\lambda_S = 80000$ CNY/h is the hourly cost of a shutdown, given in the problem statement.

Second, if we keep the intake open while the water is contaminated, we face water quality risks. The cost of this risk depends on how much the concentration exceeds the safe level (not the total concentration) and for how long:

$$C_R(u(t), C(t)) = \lambda_R \int_0^T \max(0, C(t) - C_{\text{std}}) u(t) dt \quad (5)$$

The term $\max(0, C(t) - C_{\text{std}})$ computes specifically the excess concentration above the safety threshold. We set a very high risk coefficient $\lambda_R = 1 \times 10^8$ CNY/(mg·L⁻¹·h) to enforce a “safety-first” approach. The safety standard is the warning threshold $C_{\text{std}} = 0.1$ mg/L.

Third, even during a shutdown, the city still needs water. We assume emergency measures are activated, providing 60% of the normal supply capacity. If demand still exceeds this emergency supply, a shortage penalty applies:

$$C_D(u(t), D(t)) = \lambda_D \int_0^T \max(0, D(t) - Q_{\text{eff}}(u(t))) dt \quad (6)$$

Here, $D(t)$ denotes urban water demand. The shortage penalty coefficient is $\lambda_D = 5000$ CNY/m³. The effective supply capacity Q_{eff} depends on the intake's status:

$$Q_{\text{eff}}(u) = \begin{cases} 0.6 Q_{\text{max}}, & u = 0 \quad (\text{intake shut down}) \\ Q_{\text{max}}, & u = 1 \quad (\text{intake operating}) \end{cases} \quad (7)$$

where $Q_{\text{max}} = 0.6 \text{ m}^3/\text{s}$ is the intake's maximum capacity.

Putting it all together, the total cost for a given operation plan is:

$$J(u(t), C(t), D(t)) = C_S(u(t)) + C_R(u(t), C(t)) + C_D(u(t), D(t)). \quad (8)$$

Because the pollutant concentration $C(t)$ is uncertain, we minimize the expected total cost to find the best strategy:

$$\min_{u(t) \in \{0,1\}} \mathbb{E} [J(u(t), C(t), D(t))]. \quad (9)$$

This theoretical setup clearly defines the trade-offs we face. However, solving for the perfect, moment-by-moment optimal control $u^*(t)$ is too complex for real-time use. So, in the next section, we take a more practical approach. We design a simple, rule-based strategy—a dual-threshold hysteresis controller—and tune its parameters to get a near-optimal, easy-to-implement shutdown rule.

6.2.2 Intake Shutdown Strategy

The model developed in Section 6.2.1 provides a theoretical benchmark for evaluating strategies, yet it is impractical for real-time operation. Water utility managers require clear, simple, and stable operating instructions. Therefore, we design a dual-threshold hysteresis controller that translates the complex dynamic optimization problem into an executable decision rule through two tunable parameters.

To realistically simulate the water supply system's dynamic response, we build a continuous water demand function using linear interpolation based on the provided 24-hour relative demand profile, assuming the average urban demand is 80% of the maximum intake capacity. During simulation, we incorporate an innovative emergency water-rationing measure: when the intake is shut down, a municipal water-conservation notice is activated, temporarily reducing urban demand to 60% of the normal level, which better reflects actual emergency management practices.

After analyzing operational requirements and engineering constraints, we simplify the control logic into two core parameters:

- **Shutdown threshold** c_{off} : the intake is shut down when the measured concentration exceeds this value.
- **Restart threshold** c_{on} : the intake is allowed to restart only when the measured concentration falls below this value and safety is confirmed.

To balance responsiveness with operational stability, we relate the two thresholds by a ratio r :

$$c_{\text{on}} = r \cdot c_{\text{off}}, \quad \text{where } 0 < r < 1.$$

We intentionally set $c_{\text{on}} < c_{\text{off}}$ to create a hysteresis band. This design effectively prevents frequent on-off switching caused by measurement noise or natural fluctuations near a single threshold, thereby maintaining stable operation of the water supply system.

The concrete decision rule is as follows:

Shutdown conditions (shut down if either is met):

1. $C_m(t) \geq c_{\text{off}}$ (reaching the optimized threshold).
2. $C_m(t) \geq 0.3 \text{ mg/L}$ (reaching the mandatory safety limit).

Restart conditions (both must be satisfied):

1. $C_m(t) \leq c_{\text{on}}$ (falling below the restart threshold).
2. $C_m(t) < 0.3 \text{ mg/L}$ (ensuring a safe margin from the mandatory limit).

We adopt this design to ensure both responsiveness to our optimized thresholds and strict adherence to regulatory safety limits. This transforms the complex control problem into a tractable parameter optimization task: finding the optimal c_{off} and r that minimize expected total cost under the baseline D=50 scenario.

6.3 Results

Based on our cost-optimization model, we developed a clear operational strategy for the water intake system. As can be seen from Figure 6, the rule is straightforward:

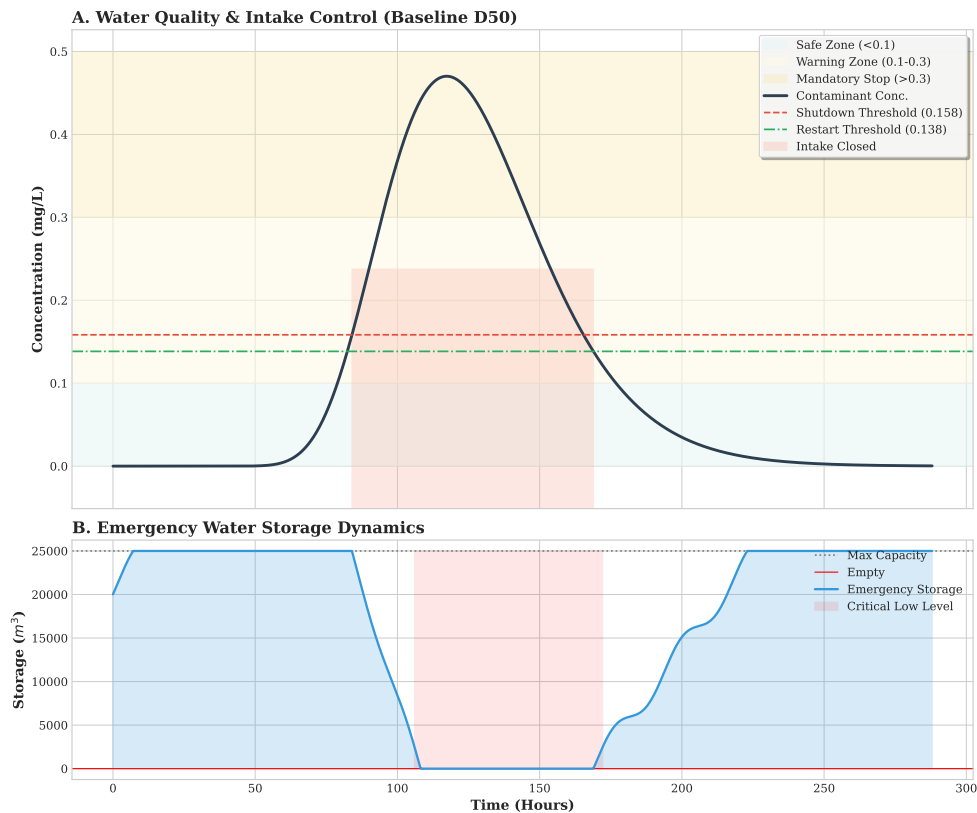


Figure 6: Water Quality Safety & Emergency Storage Dynamic Response

- Shut down the intake when pollutant concentration reaches 0.1584 mg/L
- Restart only when concentration drops below 0.1384 mg/L and remains under the 0.3 mg/L safety limit

This strategy maintains a 0.02 mg/L gap between shutdown and restart thresholds to prevent frequent switching. The shutdown level is intentionally set below the regulatory limit to ensure a safety buffer.

We tested this strategy under $D = 50$ pollution scenario. The results show:

The proposed strategy successfully maintains pollutant concentrations below the 0.3 mg/L safety limit throughout the entire event. However, despite activating 60% emergency water rationing measures, the system still experiences a water shortage of 63,558 m³. The intake remains shut down for 84.9 hours, representing 50.5% of the total simulation period, while the total mass of excess pollutants ingested is limited to only 0.6267 kg.

Our analysis reveals that complete safety requires accepting some water shortage. The current 25,000 m³ reservoir cannot fully buffer a multi-day pollution event, resulting in significant unmet demand even with an optimal strategy.

This outcome does not indicate strategy failure but rather identifies system limitations. The 0.1584/0.1384 threshold pair represents the best possible compromise under current constraints, prioritizing public health while minimizing disruption.

The results suggest that infrastructure improvements—such as reservoir expansion or water-blending technology—would be necessary to achieve both perfect safety and continuous supply during extended pollution events.

6.4 Robustness Analysis

Based on the model and optimization established in previous sections, we tested our strategy under three typical pollution scenarios ($D = 10$, $D = 50$, $D = 200$). The results show that the pollutant concentration remained below the safety limit in all scenarios, verifying the robustness of the 0.1584/0.1384 mg/L dual-threshold rule against uncertainty.

Meanwhile, the increasing trend of water shortage from 32,809 m³ ($D = 10$) to 81,019 m³ ($D = 200$) indicates that, under the current infrastructure constraints, a trade-off inevitably exists between water safety and supply continuity. Our strategy provides a scientifically optimized balance between the two by means of threshold tuning.

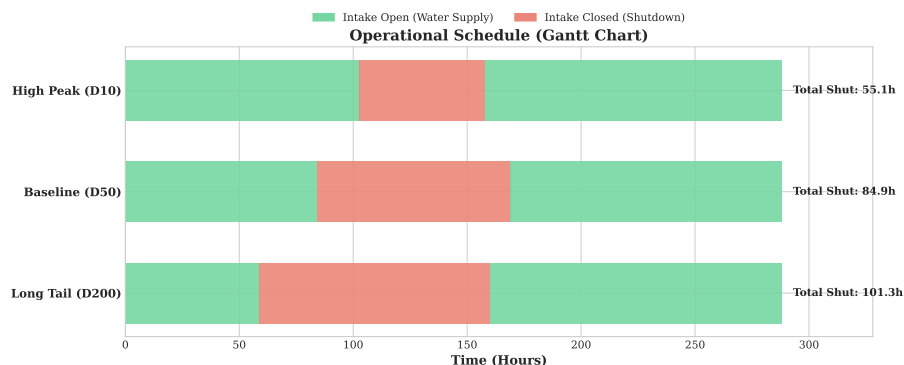


Figure 7: Water Intake Operational Schedule: Dispersion Coefficient Scenarios

By analyzing the operational time allocation under different scenarios (Figure 7), we observe that the strategy can adapt its response intensity intelligently: shutdown duration increases reasonably from 55.1 h ($D = 10$) to 101.3 h ($D = 200$). This demonstrates the rule's ability to adjust automatically according to pollution characteristics, which enhances practical operability.

7 Model III: Sensor Network

7.1 Model Overview

We built a sensor-placement optimization model that aims to get the most early warning time possible with a limited number of sensors. Our method combines a multi-objective scoring function with a two-layer robust optimization algorithm. The model accounts for practical factors like measurement noise, reporting delays, spatial coverage, and detection reliability. Using a greedy selection process inside and Monte Carlo validation outside, we solved for the best sensor layout. The results show that the optimized network (3, 8, and 18 km) reduces the time to first detection from 77.18 hours to just 2.37 hours, proving that the algorithm works well even when there is uncertainty.

7.2 Model Establishment

We have a limited budget of at most three online concentration sensors and need to choose the best layout from 11 candidate locations (3, 5, 8, 10, 12, 15, 18, 20, 22, 25, 28 km). The sensors have 0.02 mg/L measurement noise and a 5-minute reporting delay. Our goal is to get as much warning time as possible before pollutants reach the water intake 30 km downstream.

To evaluate the candidate locations thoroughly, we designed a scoring function that combines five key factors:

- Warning-time gain (ΔT): how much earlier we can detect compared to the baseline
- Signal-to-noise ratio (SNR): ability to withstand measurement noise
- Detection-window length (W): total time the concentration stays above the 0.1 mg/L threshold
- Spatial-correlation penalty ($1 - \rho$): encourages sensors to be spread out rather than clustered
- Detection reliability (R): false-alarm resistance evaluated through random simulation

The overall score is calculated as:

$$\text{Score} = \Delta T \times \text{SNR} \times W \times (1 - \rho) \times R, \quad (10)$$

meaning that a weakness in any one factor will noticeably lower the total score.

We built a two-layer robust optimization algorithm.

The inner layer uses a greedy selection: starting with an empty set, it picks the sensor with the highest current score each time, then updates the baseline detection time.

The outer layer performs Monte Carlo validation: multiple independent runs simulate different noise conditions, we count how often each location is chosen, and finally select the three most-frequently chosen sites as our sensor network.

This approach ties multi-objective optimization to practical engineering. It avoids the computational burden of checking every possible combination while still delivering a high-quality, stable solution. By using frequency-based selection, our design remains reliable even when measurement noise is present.

7.3 Results and Robustness Analysis

7.3.1 Optimal Layout Determination

Through our robust optimization algorithm, we determined the optimal sensor network consisting of three locations:

- **3 km:** Core near-field early-warning point
- **8 km:** Mid-field verification point
- **18 km:** Far-field strategic sentinel

We constructed a tiered monitoring system. The 3 km position provides the earliest warning, the 8 km position confirms pollution trends, and the 18 km position offers the maximum warning lead-time. This distribution ensures comprehensive coverage from near-field to far-field.

7.3.2 Performance Improvement Quantification

The optimized network achieves a breakthrough in warning time:

- Baseline detection time (intake only): 77.18 hours
- Optimized network detection time: 2.37 hours
- Warning-time gain: +74.81 hours (equivalent to 3.12 days earlier warning)

The 74.8-hour warning window enables a scientific verification process: ‘the initial alarm → secondary confirmation → final decision’ three-step mechanism effectively avoids unnecessary shutdowns caused by false alarms.

In terms of emergency preparedness, we can optimize reservoir dispatch 74.8 hours in advance to maximize the utilization of the 25,000 m³ storage capacity; have sufficient time to activate backup water sources or coordinate external water transfers; and gradually issue public notifications to avoid social panic.

7.3.3 Stability Analysis

Results from 100 independent Monte Carlo runs show:

The 3 km location was selected in all 100 Monte Carlo runs (confirming its irreplaceable role as the core early-warning point). The 8 km (26% selection frequency) and 18 km (25% selection frequency) sites further balance spatial coverage: 3 km provides the earliest possible detection, 8 km verifies emerging pollution trends and 18 km maximizes the warning lead-time before pollutants reach the downstream intake.

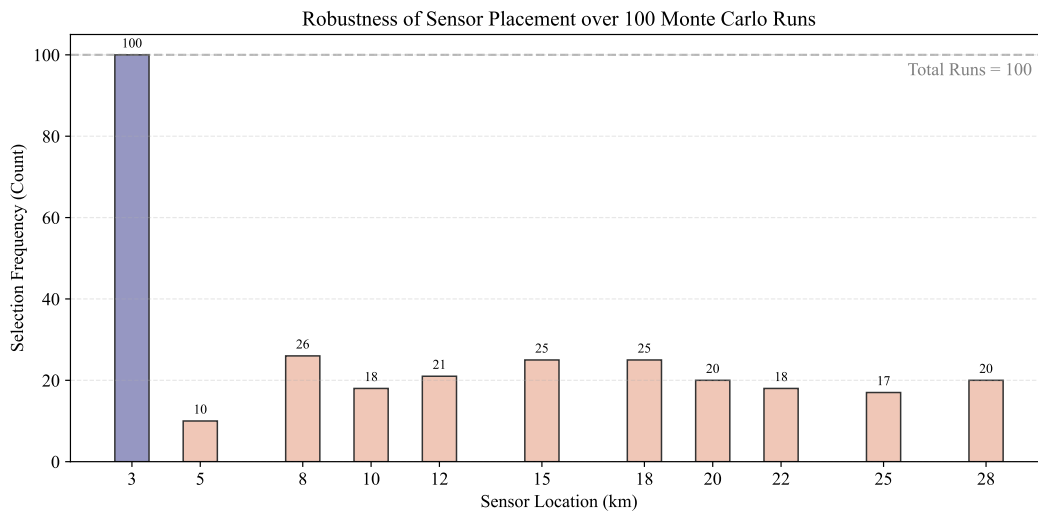


Figure 8: Selection Frequency of Candidate Sensor Locations Across 100 Monte Carlo Runs

As shown in Figure 8, this tiered layout (3–8–18 km) ensures comprehensive monitoring from near-field to far-field, addressing both timeliness and reliability of pollution detection.

8 Sensitivity Analysis

To verify the reliability and assess the parameter dependence of our primary one-dimensional advection-dispersion-reaction (ADR) model, we conducted a sensitivity analysis on the key parameters controlling pollutant transport to assess their impact on arrival time, peak concentration, and threshold-exceedance duration at the intake.

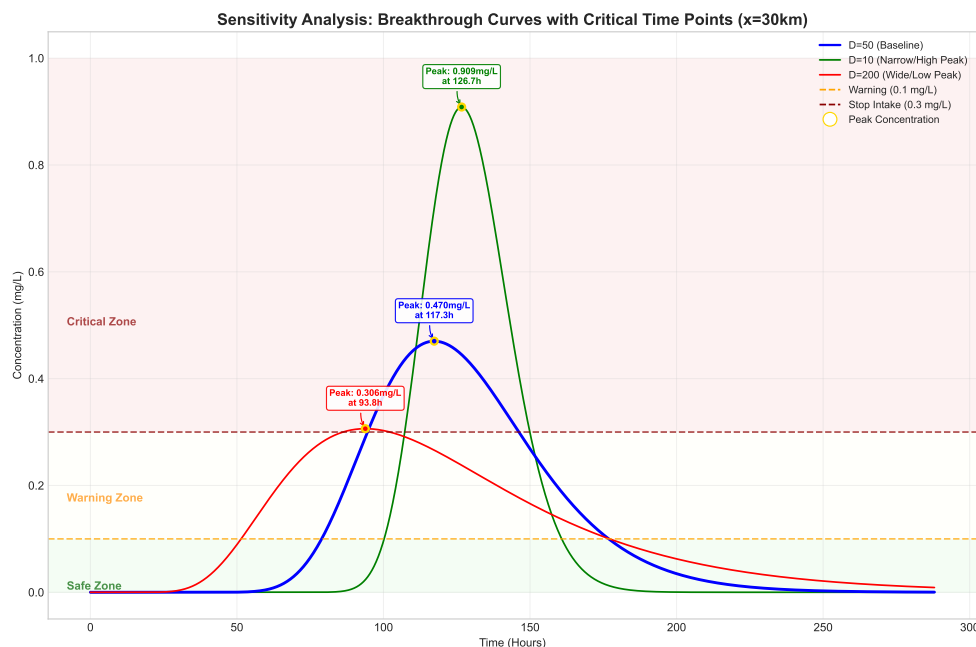


Figure 9: 30 km Breakthrough Curves: Effect of Dispersion Coefficient (D)

The dispersion coefficient D has the strongest effect. As Figure 9 shows, increasing D from 10 to 200 m^2/s makes the peak concentration drop by about two-thirds—from 0.91 mg/L down

to 0.31 mg/L. However, the time the intake stays above the 0.1 mg/L warning level increases significantly. We also notice that above the higher 0.3 mg/L shutdown level, the exposure time is longest at $D = 50 \text{ m}^2/\text{s}$, not at the highest D . This tells us that stronger dispersion spreads out the pollution: the peak gets lower, but lower concentrations can last longer, and sometimes the highest threshold is not even reached.

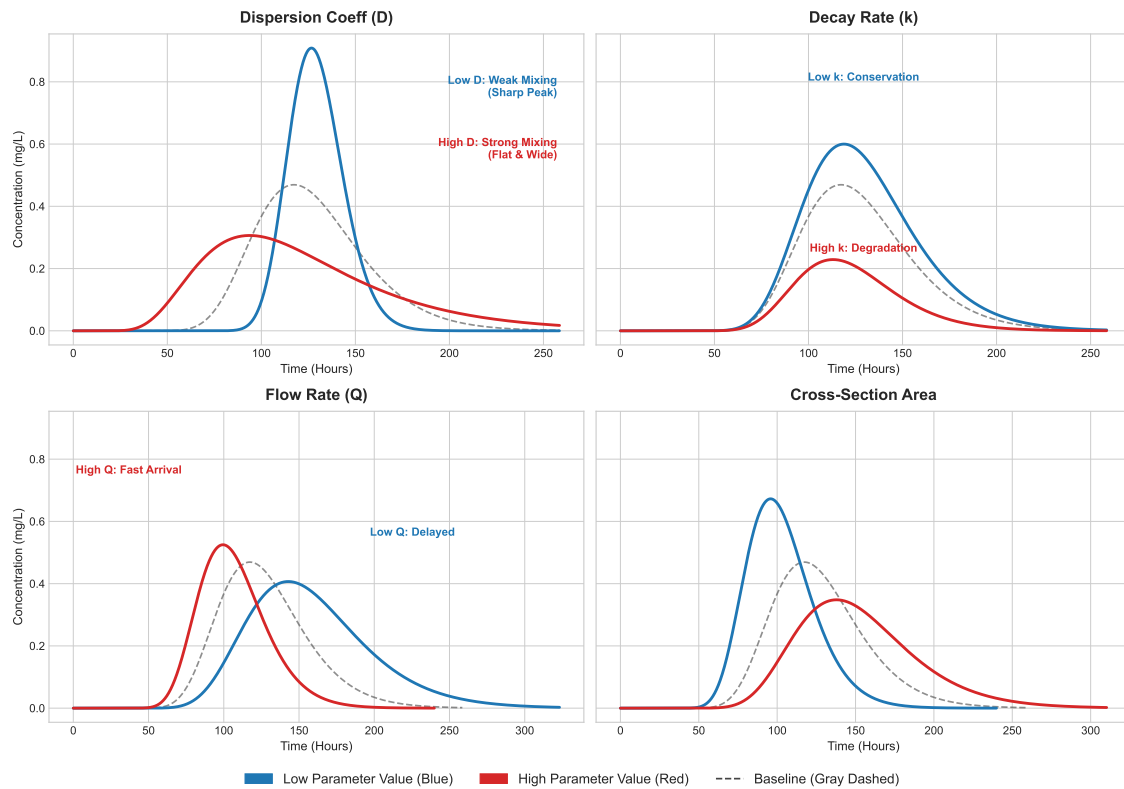


Figure 10: Sensitivity Analysis

Sensitivity analysis was also performed on three additional parameters: decay rate (k), flow rate (Q), and cross-section area (A), with results presented in Figure 10.

Our analysis employs a decay rate of $k = 0.05 \text{ day}^{-1}$ as the baseline. Sensitivity analysis reveals that even under a more conservative zero-decay assumption ($k = 0$), the variations in key metrics such as peak concentration and exceedance duration remain minor. This confirms the robustness of the model predictions with respect to the decay rate parameter. Consequently, the baseline results derived using $k = 0.05 \text{ day}^{-1}$ provide a sufficient and reliable foundation for safe operational planning.

Flow rate (Q): Dominantly affects pollutant arrival time—higher Q accelerates advection, leading to earlier peak arrival, while peak concentration and exceedance duration show little variation;

Cross-section area (A): Larger A reduces velocity, resulting in delayed arrival time and slightly extended exceedance duration, but has no significant impact on peak concentration.

Accurate forecasting of Q (and thus v) is critical for ensuring sufficient early warning time, even though these parameters do not strongly affect peak concentration magnitudes.

9 Strength and Weakness

9.1 Strength

- **Established a complete response loop from early warning to decision-making**

We integrated physical modeling of pollutant transport, economic-risk trade-off optimization, and sensor placement under engineering constraints into a collaborative system.

This framework follows the logical chain of “sensor network (early detection) → diffusion model (concentration prediction) → dual-threshold controller (scientific decision) → sufficient warning window (smooth implementation)”.

This integrated design ensures theoretical consistency and engineering feasibility, providing a systematic solution for responding to sudden water pollution incidents.

- **Developed key models that balance rigor and practicality**

In the decision-making model, we transformed a complex optimal control problem into an easy-to-operate “dual-threshold hysteresis controller”, with its optimal parameters determined through mathematical optimization. In the sensor network, we employed a multiplicative scoring function that integrates multiple indicators for layout optimization, ensuring the reliability and robustness of early warning.

- **Conducted thorough sensitivity analysis and scenario validation**

We systematically examined the sensitivity of the transport model to key parameters within reasonable ranges and verified the effectiveness of the shutdown strategy under multiple typical pollution scenarios. The results show that the core strategy remains stable under different hydrological and pollution conditions, demonstrating reliability for practical application.

9.2 Limitations

- **Necessary simplification of physical processes**

The one-dimensional transport model performs well in predicting overall trends over long river reaches but has inherent limitations in depicting mixing details caused by local complex hydraulic structures.

- **Adaptability constraints of steady-state parameters**

The model is based on fixed cross-sectional geometry and steady hydrological inputs, and its ability to dynamically update in response to drastic geomorphic scouring or flow surges caused by extreme events needs enhancement.

10 Further Discussion

Our model has room for further optimization in practical application. A natural improvement direction is to make the system more intelligent—by establishing a data assimilation mechanism that dynamically feeds real-time monitoring data from sensors back into the model, enabling continuous self-correction during prediction and transforming it from a static computational tool into an adaptive system with learning capabilities.

In practice, this framework can be developed into two types of practical tools. For individual water treatment plants, it can serve as an intelligent assistant for operational decision-making, providing standardized risk assessments and operational guidance. For a river basin or city, it can be extended into a comprehensive emergency management platform, integrating multi-source information such as meteorology and hydrology to assist managers in conducting whole-process simulation and coordinated response. In regions with complex water source conditions or higher risks, such tools are particularly valuable, making water supply security more systematic and scientific.

From a research perspective, two directions are worth exploring: first, improving pollutant diffusion prediction through data-driven methods to enhance simulation efficiency under complex conditions; second, integrating ecological and social dimensions into the decision-making framework to make risk management more comprehensive. These explorations will advance water pollution emergency response from passive reaction toward intelligent prevention and control.

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Appendices

Appendix A First appendix

Here are some additional visual results from our work.

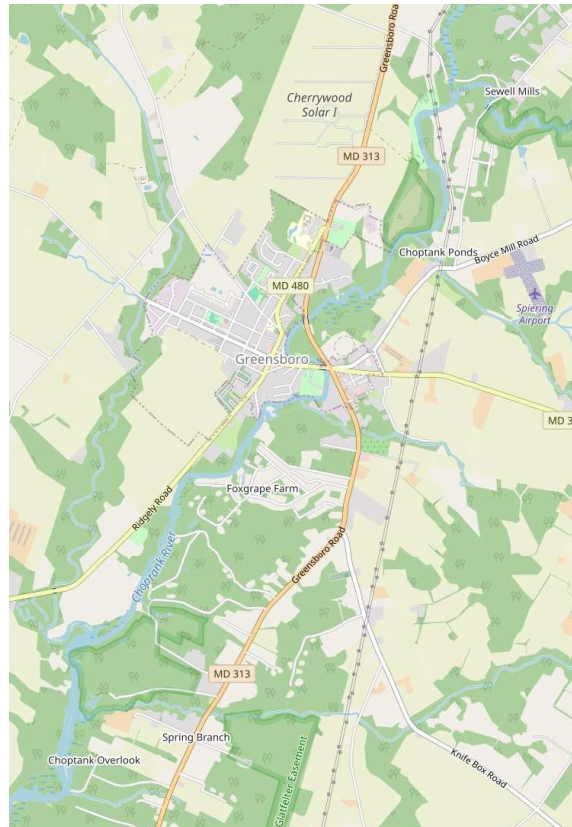


Figure 11: Location of USGS gauge 01491000

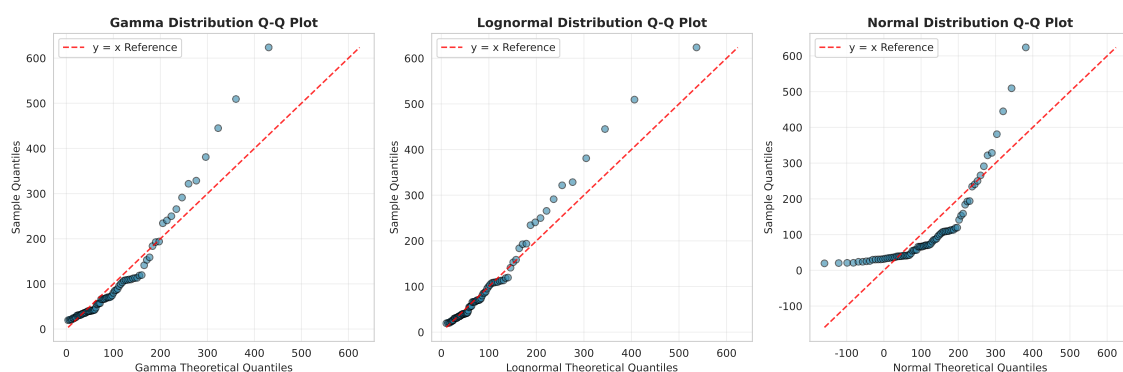


Figure 12: June Streamflow Distribution Q-Q Plot

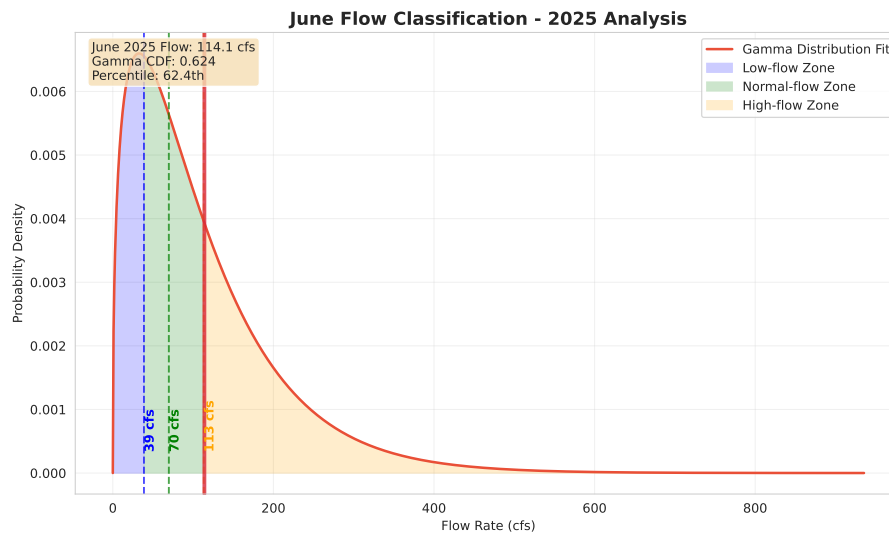


Figure 13: June Flow Classification

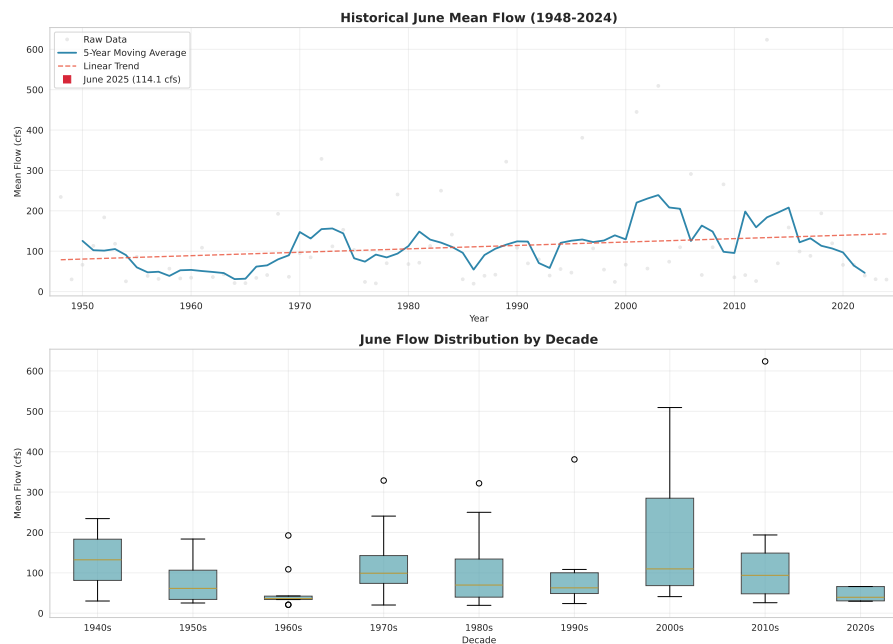


Figure 14: Historical June Streamflow

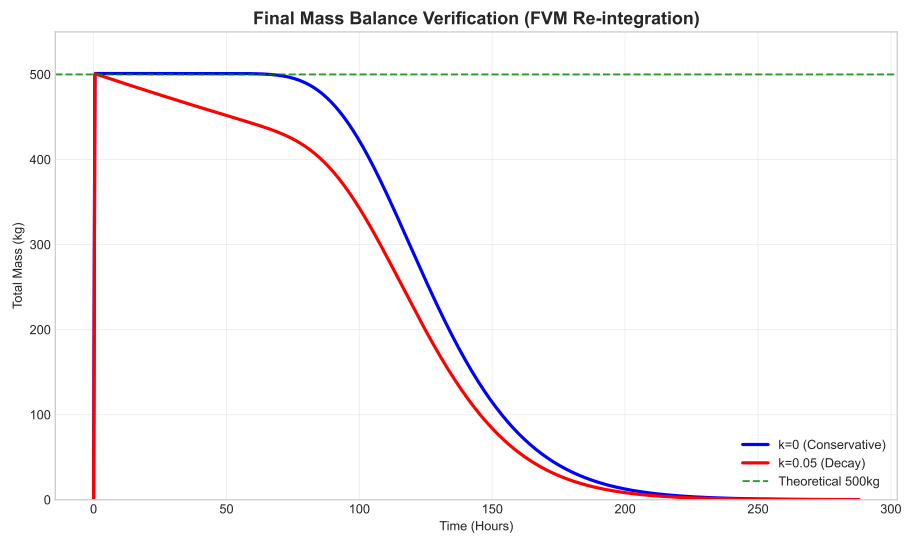
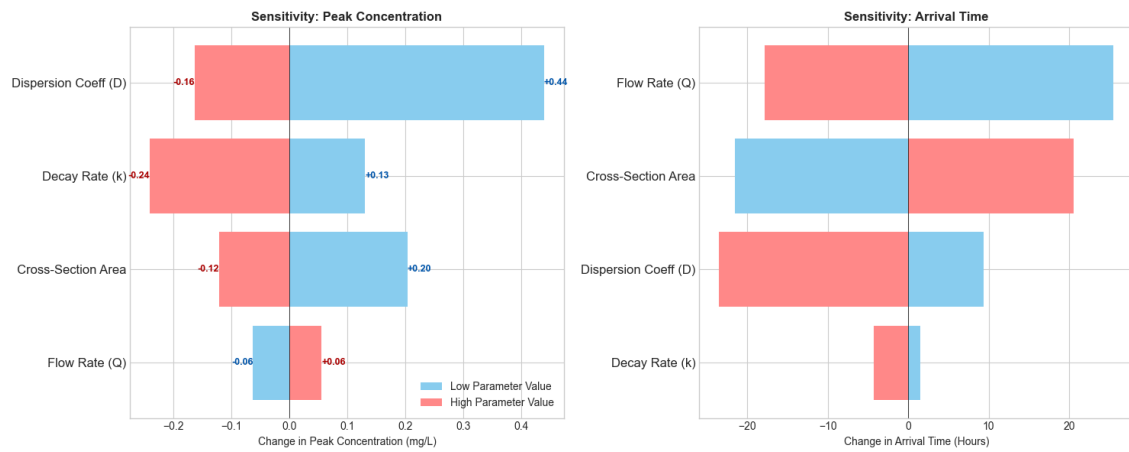


Figure 15: Mass Balance Verification ($k = 0$ v.s. $k = 0.05$)



Appendix B Second appendix

Key code segments are provided below:

Algorithm 1: Water Intake Operation Logic (Dual-Threshold Hysteresis Control)

Input : Concentration time series $c_{\text{river}}(t)$, thresholds c_{off} , c_{on} , C_{mand} , C_{safe} , maximum intake capacity Q_{max} , time step Δt .

Output Intake status $u(t)$, total ingested risk mass M_{risk} , violation duration T_{viol} .

:

Initialize $u \leftarrow 1$ (open), $M_{\text{risk}} \leftarrow 0$, $T_{\text{viol}} \leftarrow 0$;

for each time step i do

$c \leftarrow c_{\text{river}}[i]$;

if $u = 1$ (intake open) then

if $c \geq c_{\text{off}}$ **or $c \geq C_{\text{mand}}$ then**

$u \leftarrow 0$ // Shut down: reaches either control or mandatory limit

else

if $c \leq c_{\text{on}}$ **and $c < C_{\text{mand}}$ then**

$u \leftarrow 1$ // Restart: below restart threshold and safe from mandatory limit

if $u = 1$ then

$Q_{\text{eff}} \leftarrow Q_{\text{max}}$ // Full capacity when open **if $c > C_{\text{safe}}$ then**

$M_{\text{risk}} \leftarrow M_{\text{risk}} + (c - C_{\text{safe}}) \times Q_{\text{max}} \times \Delta t$ // Accumulate health risk mass

if $c > C_{\text{mand}}$ then

$T_{\text{viol}} \leftarrow T_{\text{viol}} + \Delta t$ // Accumulate mandatory violation time

else

$Q_{\text{eff}} \leftarrow Q_{\text{max}} \times \alpha_{\text{emergency}}$ // Reduced capacity during shutdown

$(\alpha_{\text{emergency}} = 0.6)$

Report on Use of AI

1. Doubao

Query1: What is greedy selection strategy?

Output: Greedy selection strategy is an algorithmic approach that makes the locally optimal choice at each step, aiming to reach a globally optimal solution (only valid for problems with greedy choice property and optimal substructure).

Query2: What is Monte Carlo simulation? How to use it? In what situations is it applicable?

Output: Monte Carlo simulation is a statistical method that uses random sampling + repeated iterative computation to estimate outcomes and probabilities for complex problems with uncertainty (no closed-form analytical solutions). How to use it: Define the problem and build a probabilistic model (identify uncertain variables and their distribution). Generate random samples for the uncertain variables. Run repeated simulations with the samples to calculate trial outcomes. Analyze the simulation results (distribution, mean, variance, probability of key outcomes). Applicable situations: Problems with high uncertainty/stochasticity (unpredictable variables). Complex models with no exact mathematical solution. Risk assessment, stochastic process analysis (e.g., finance risk, engineering design, project scheduling, supply chain optimization, quantum mechanics, inventory management).

Query 3: Could you recommend a color palette with low saturation for graphic design? Please provide the RGB hex codes like #FFFFFF for me.

Output: Low-saturation soft color palette (suitable for graphic design, mild contrast & high adaptability)

Main tones

#E8E2D9 (warm beige) | #D4D9DF (cool light gray blue) | #D9D6D0 (neutral oatmeal)

Accent tones

#BFB9B0 (taupe gray) | #C4C8CB (ash blue) | #D1CDC7 (muted mauve gray)

Neutral tones

#F5F3F0 (off-white) | #86888A (soft dark gray) | #6A6C6E (deep muted gray)

2. OpenAI ChatGPT (Jan 23, 2026 version, GPT-5.2)

Query 1: Where is the USGS gauging station 01491000?

Output: USGS gauging station 01491000 is located on the Choptank River near Greensboro, in the state of Maryland, United States.

Station name: Choptank River near Greensboro, MD.

River: Choptank River — a tributary flowing into Chesapeake Bay.

Coordinates: approximately 38°59'50" N latitude, 75°47'10" W longitude.

County: Caroline County, Maryland.

This USGS station monitors streamflow and water-quality data for the Choptank River at this location.